1. PROJECT TITLE

Genome-wide Identification and Structural Docking Analysis of Salt-Stress Responsive Proteins in Buckwheat (Fagopyrum esculentum) Using Yoga/Ayurvedic Phytochemicals

2. Proposed Area & Focused Topic

Proposed Area:

Plant salt stress tolerance

Focused Topic:

Genome-wide identification and structural analysis of salt-stress related SOS proteins in buckwheat

3. Literature Table (12 Papers)

NO	TITLE	YEAR	PLANT	AUTHORS	METHODOLOGY	KEY FINDINGS	RELEVANCE TO PRESENT STUDY	PROTEIN/PATHWAY TYPE	CATEGORY
01.	IN SILICO ANALYSIS AND HOMOLOGY MODELLING OF ANTIOXIDANT PROTEINS OF SPINACH	2010	SPINACIA OLERACEA (SPINACH)	Archna Sahay et al	Physiochemical characterization, homology modelling and structural validation of antioxidant proteins using computational tools.	The study successfully generated reliable three-dimensional models of antioxidant stress-responsive proteins and validated them using standard structural assessment parameters.	This work supports the feasibility and reliability of homology modelling approaches for plant stress-responsive proteins prior to molecular docking studies.	Antioxidant stress-responsive proteins	Protein Analysis And Structural Modelling.
02.	Halophytes as a source of bioactive compounds for anticancer drug discovery	2023	Multiple Halophyte species	Hamdoon A. Mohammed et al	Comprehensive literature review of phytochemical composition and biological activities of halophytes.	Halophytic plants accumulate high levels of phenolics and flavonoids under salt stress, exhibiting strong antioxidant and bioactive properties.	The findings justify the selection of phytochemicals from stress-adapted plants for docking with salt-stress adapted plants for docking with salt-stress responsive proteins.	Secondary metabolites	Phyto chemicals and bioactive compounds.
03.	A comprehensive review of transcription factor-mediated in plants under	2025	General plant systems	Karim Rabeh et al	Review of transcription factor regulation of secondary metabolite biosynthesis under stress conditions.	Stress-responsive Transcription Factors regulate Secondary Metabolite	Provides conceptual linkage between salt stress signaling pathways and	Transcription factors	Regulatory Proteins and Stress Response.

	environmental stress					Accumulation during salinity and other Environmental stresses.	phytochemical accumulation, supporting phytochemical docking rationale.		
04.	Genome-wide Identification of MAPK Gene Family members in Fagopyrum Tataricum and Their expression during development and stress responses.	2022	Tartary Buckwheat	Yingjun Yao et al	Genome-wide identification, phylogenetic analysis, gene expression profiling, and salt stress experiments.	16 MAPK genes were identified; FtMAPK1 enhanced antioxidant activity and regulated stress-related gene expression under salt stress.	Demonstrates the role of MAPK signaling proteins in buckwheat salt stress tolerance and supports their inclusion as candidate proteins.	MAPK signaling proteins	Stress signaling proteins
05.	Salt Tolerance Mechanisms of Plants	2020	Model plants and halophytes	Eva van Zelm et al	Review of physiological and molecular salt tolerance mechanisms.	Salt stress activates ROS signaling, ion transporters, hormonal pathways, and adaptive growth responses.	Provides a foundational understanding of salt stress mechanisms for interpreting genome-wide protein identification results.	General salt stress pathways	Stress physiology review
06.	Mechanism of Salinity Tolerance in Plants: Physiological, Biochemical, and Molecular Characterization	2014	Model plants and crops	Bhaskar Gupta et al	Review of plant responses to salinity at multiple biological levels.	Plants maintain ion homeostasis and osmotic balance through antioxidants, hormones, and metabolic adjustments.	Guides functional annotation and interpretation of salt stress-responsive proteins in buckwheat.	General stress response	Stress physiology review
07.	Unraveling Salt Stress Signaling in Plants	2018	Model plants	Yongqing Yang et al	Review of genetic and biochemical signaling pathways under salt stress.	MAPK cascades and SOS signaling play central roles in ion homeostasis and stress adaptation.	Supports targeting SOS and MAPK proteins for genome-wide identification and docking.	MAPK and SOS signaling	Stress Signaling pathways
08.	The Classical SOS Pathway Confers Natural Variation of Salt Tolerance in Maize	2022	Zea mays	Xueyan Zhou et al	Genetic and molecular analysis of SOS pathway components.	SOS1 and associated regulators control Na^+ efflux and salt tolerance variation.	Provides comparative insights for identifying SOS pathway homologs in buckwheat.	SOS signaling and ion transporters	Ion transport And Stress response
09.	Nonselective Cation Channels (NSCCs) in Plants	2007	General plants	M. Maathuis et al	Review of NSCC families and physiological roles	NSCCs regulate Na^+, K^+, and Ca^{2+} flux under salinity stress.	Helps identify ion channel proteins for modeling and docking.	Ion channels	Ion transport and stress response
10.	Genome-wide identification of GRAS gene family in Tartary buckwheat	2019	Tartary buckwheat	Moyang Liu et al	Genome-wide analysis, motif analysis, expression profiling.	Forty-seven GRAS genes identified with stress-responsive expression patterns.	Supports transcription factor involvement in salt stress and metabolite regulation.	GRAS transcription factors	Transcription factors
11.	Genome-wide Investigation of Hsf Gene Family in Tartary Buckwheat	2019	Tartary buckwheat	Moyang Liu et al	Genome-wide identification and expression analysis.	Hsf genes regulate abiotic stress responses and development.	Provides candidate stress-responsive regulatory proteins.	Heat shock factors	Transcription Factor
12.	Genome-wide Identification of SPL Gene Family	2019	Tartary buckwheat	Moyang Liu et al	Genome-wide analysis and expression profiling.	SPL genes are involved in growth,	Supports inclusion of SPL proteins in	SPL transcription factors	Transcription factors

	in Tartary Buckwheat					development, and potential stress signaling.	stress-response analysis.		

7. Grouped Literature Analysis

1. Salt stress basics: Papers 5, 6, 7- General salt stress mechanisms, ROS, ABA/auxin signaling, ion homeostasis, adaptive responses.

2. SOS pathway & ion transport: Papers 4, 8, 9- SOS signaling, MAPK cascades, NSCC ion channels; guides genome-wide identification and docking targets in buckwheat.

3. Genome-wide studies in buckwheat: Papers 10, 11, 12- Genome-wide identification of TFs (GRAS, Hsf, SPL); candidate proteins for stress response & growth.

4. Structural & docking studies: Papers 1- Homology modeling and structure validation of stress-responsive proteins; guides computational docking.

5. Phytochemicals & stress modulation: Papers 2, 3- Bioactive compounds from medicinal and stress-adapted plants; rationale for docking with target proteins.

8. Literature Summary

Salt stress is one of the most severe abiotic stresses affecting plant growth and crop productivity worldwide. Studies in model plants and major crops have shown that salinity disrupts ion homeostasis, induces osmotic stress, and leads to excessive accumulation of reactive oxygen species (ROS), which negatively affect cellular metabolism and development (Gupta et al., 2014; van Zelm et al., 2020). To cope with these effects, plants activate a range of physiological and molecular responses, including antioxidant defense systems, ion transport mechanisms, hormonal regulation, and stress-responsive gene networks (Yang et al., 2018).

At the molecular level, salt stress signaling is largely mediated by the Salt Overly Sensitive (SOS) pathway, MAPK signaling cascades, and ion transport systems. The SOS pathway, composed of SOS1, SOS2, and SOS3, plays a central role in maintaining sodium ion homeostasis through Na^+ extrusion under saline conditions. Genetic studies in crops such as maize have demonstrated that variation in SOS pathway components significantly influences salt tolerance (Zhou et al., 2022). In addition, nonselective cation channels (NSCCs) regulate Na^+, K^+, and Ca^{2+} fluxes during salt stress and contribute to ionic balance and stress signaling (Maathuis et al., 2007). MAPK cascades further function as key regulators that link stress perception with transcriptional control and antioxidant responses (Yang et al., 2018).

Buckwheat (*Fagopyrum* spp.) is a nutritionally important pseudo-cereal known for its adaptability to adverse environmental conditions. Genome-wide studies in Tartary buckwheat (*Fagopyrum tataricum*) have identified several stress-responsive gene families, including MAPKs, GRAS, Hsf, and SPL transcription factors. Yao et al. (2022) reported differential expression of MAPK genes under salt stress, with certain members enhancing antioxidant activity and regulating stress-related gene expression. Similarly, GRAS, Hsf, and SPL transcription factors have been reported to participate in abiotic stress responses and developmental regulation in buckwheat (Liu et al., 2019).

Computational approaches have been widely used to study plant stress-responsive proteins, particularly when experimental structures are unavailable. Homology modeling and in silico structural validation have been successfully applied to antioxidant and stress-related proteins, enabling insights into protein structure, stability, and function (Sahay et al., 2010). These structural analyses form the basis for molecular docking and molecular dynamics simulations to study protein–ligand interactions.

In addition, phytochemicals derived from halophytic and medicinal plants have been extensively reported for their antioxidant and protective properties under stress conditions. Such plants accumulate phenolics, flavonoids, and other secondary metabolites during salinity stress, which contribute to stress tolerance mechanisms (Mohammed et al., 2023). Stress-responsive transcription factors have also been shown to regulate secondary metabolite biosynthesis during salinity stress, indicating a close association between stress signaling pathways and phytochemical production (Rabeh et al., 2025).